DeadEnd

The reasoning behind the code.

CodersInPune 2026 | Team NIKKUSHRIVASTAVA0 | Lead: Nitanshu Shrivastav

LIVE PRODUCT
deadend-ai-codersinpune.vercel.app

SOURCE CODE
github.com/nikkushrivastava0-a11y/deadend-ai

The problem we chose

When a developer leaves, the repository and issue tracker survive, but the reasons for rejected technical approaches may remain buried in discussions. Incoming developers risk repeating an expensive experiment that has already failed.

DeadEnd preserves the negative knowledge: what was attempted, why it was abandoned, how strong the evidence is, and what should be checked before trying again. It turns this information into a human-reviewed handover with original sources intact.

Verified production release: 10 November 2026, commit **9ff5e302e176**. Full engineering test suite: **23/23 passed**. Public production import, correction, approval, evidence citation and handover were checked through HTTP. Interactive browser testing on a physical phone remains unverified.

A five-step handover, end to end

01 / Connect

Import a public GitHub issue or pull-request discussion, or paste synthetic/authorised notes. Keep source text and its original permalink.

02 / Reconstruct

Recover explicit rejected experiments, blockers and failures. When a comment says “We reverted it”, label same-comment context without changing the evidence excerpt.

03 / Verify

Check the exact quotation against the original comment segment. Flag missing reasoning and possible later updates rather than guessing.

04 / Review

Edit the interpretation, approve, dispute, reject or reopen a finding. Preserve its original quotation and source link.

05 / Handover

Generate a Markdown continuation brief containing only approved public findings, supporting evidence and next actions.

How a judge can try it

Open the example case, inspect Stripe and its possible later update, edit the interpretation, approve the public finding and download the handover. Alternatively, use Add Evidence to import an actual public GitHub issue. The example is fictional and clearly labelled.

Implemented architecture and technical choices

Component	Shipped implementation
Interface	Responsive vanilla JavaScript, semantic HTML and interactive CSS 3D evidence constellation. Pointer, keyboard and reduced-motion/compact fallbacks.
API and integrations	Node.js HTTP API with Vercel adapter; selected public GitHub issues, pull requests, issue comments, and inline reviews.
Extraction	Rule-based explicit failure/rejection language; context-aware markdown approach heading; distinct per-segment evidence identifiers.
Verification and review	Exact excerpt and original segment checks, source permalinks, human corrected interpretation and review state.
Public handover	Only approved public claims. Markdown export carries quote, source link, next action and cautionary related updates.
Session handling	Time-limited AES-256-GCM browser-held workspace capsules; sessions isolated, not durable team accounts.
Model path	Optional structured model excerpts with exact quote verification; no live model credential configured in production as checked.

Changes from the Round 1 proposal

The original technical plan proposed Next.js, Supabase and an OpenAI Agents SDK. To complete the core workflow within the event, we instead used a dependency-free Node server and vanilla frontend, and CSS perspective rather than WebGL. Full enterprise permission controls and multi-source AI reconciliation were deferred. These changes prioritised an evidence-linked, testable MVP over integration breadth.

Real public evidence and tested results

Non-mocked public GitHub case

tinyhumansai/openhuman issue #867 documents a series of abandoned webview-loading approaches. The public production import returned HTTP 200 with three candidate findings, including two named approaches. Editing and approving a finding produced a handover with the exact original evidence excerpt and source URL.

Additional public case verification

Ruuvi issue #501 describes a historical comparison-chart idea that was abandoned, followed by a possible future reconsideration. DeadEnd labels the latter as a review lead rather than proof the old limitation has disappeared.

CodeAtlas issue #682 describes a reverted Kanban substitution whose single rejection sentence refers back to a previous sentence. The release extracts that context while retaining the exact unchanged evidence quote.

Verified engineering checks

Check	Observed result
Automated tests	23/23 passing on verified candidate commit 904bb1a; GitHub Actions successful.
Release	PR #1 merged to main as 9ff5e302; Vercel production READY.
Public endpoints	/ and /api/health returned HTTP 200; deployed existing report PDF responded 200 as application/pdf.
Synthetic path	Five findings, Stripe correction, approved handover. No restricted sample findings exported.
Real-data path	Production GitHub import #867: 3 findings. Exact quotation and original URL survived approval and handover.

Limitations we will not hide

• Production model configuration currently reports aiConfigured=false. The working extraction path uses deterministic rules; no live model-assisted extraction was claimed.

• No individual login identities, server-side role-based authorisation, enterprise private-source permissions, or Supabase RLS. The demo is for synthetic or public authorised data only.

• Workspaces are anonymous and expire; they do not support durable multi-user organisational collaboration.

• GitHub API page limits, upstream errors and missing rejection language can result in incomplete or zero findings; this is not automatic proof of no project risks.

• Heuristic related-update hints can be wrong; they are never treated as independently verified reversals.

• 3D uses CSS transforms, not WebGL/Three.js. Static responsive checks used emulation; full interaction on a physical mobile device remains unverified.

Security posture

Restricted sample entries cannot be approved into a public handover. Public sources retain original quotations and links; reviewers may correct interpretation without rewriting the evidence. This is a hackathon prototype, not a safe destination for private credentials or confidential organisational uploads.

Exactly three Round 3 commitments

01 / Decision revalidation workflow

Introduce owner-triggered recheck requests, dated status history and a clear current constraint state so an incoming developer can decide whether the original rejection still applies.

02 / Cross-source disagreement resolver

Present conflicting claims from at least two authorised sources, preserve both supporting excerpts and require an owner to resolve the disagreement before approval.

03 / Role-aware handover with receipt

Add authenticated users, server-enforced source visibility, versioned approved handovers and recipient acknowledgement so knowledge is shared only with authorised people.

People, tools and integrity

Team lead Nitanshu Shrivastav coordinated solution direction, engineering and release. Kanishk was assigned frontend/UI-UX responsibility; actual individual contributions should be reconciled with GitHub history. AI assistance contributed to research, code and documentation, and is disclosed. All public-case tests used actual authorised public sources; synthetic Stripe records are explicitly fictional.

Submission readiness

Only the team leader should submit Round 2. Before submitting, verify the live site, public repository and final shared PDF in an incognito browser. Form submission was not performed automatically.

Live: deadend-ai-codersinpune.vercel.app
Code: nikkushrivastava0-a11y/deadend-ai
Release PR: GitHub PR #1